



INTRODUCTION TO ARTIFICIAL INTELLIGENCE IN HEALTHCARE

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During This Course, You Will Learn

- The basic ideas behind Artificial Intelligence (AI) and Machine Learning (ML)
- Common terms and concepts used in AI/ML
- Different types of medical data used in AI applications
- Benefits and challenges of integrating AI in practice

Upcoming Lectures

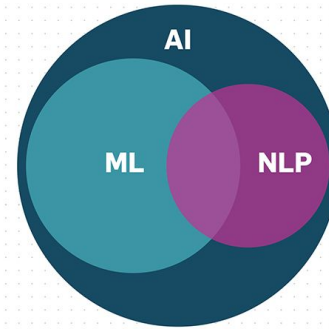
- **Introduction to AI and How Data Shapes Modern ML.**
- Overview of Machine Learning and Deep Learning Algorithms.
- LLMs Are Awesome!
- How to read an AI paper and evaluate an AI product.
- AI Applications in Medicine and Current Research at the University of Alberta.

Artificial Intelligence in Our Life



Artificial Intelligence

- Artificial intelligence (AI) are machines that can mimic the cognitive functions of humans to perform tasks of problem-solving and learning.



Artificial Intelligence

When a machine is able to mimic human intelligence by having the ability to predict, classify, learn, plan, reason and/or perceive.

Machine Learning

A subset of AI that incorporates math and statistics in order to learn from the data itself, and improve with experience.

Natural Language Processing

Subset of AI that helps systems to derive meaning and understanding from language.



Brief History of AI

- 1943- McCulloch and Pitts propose the first model of artificial neurons.
- 1950- Alan Turing introduces the Turing test.
- 1956- Term “Artificial Intelligence” coined at the Dartmouth Conference.
- 1980s- Rise of expert systems like DENDRAL.
- 1997- IBM’s Deep Blue defeats chess champion, Garry Kasparov.
- 2000s- Advances in machine learning and neural networks fuel modern AI applications.

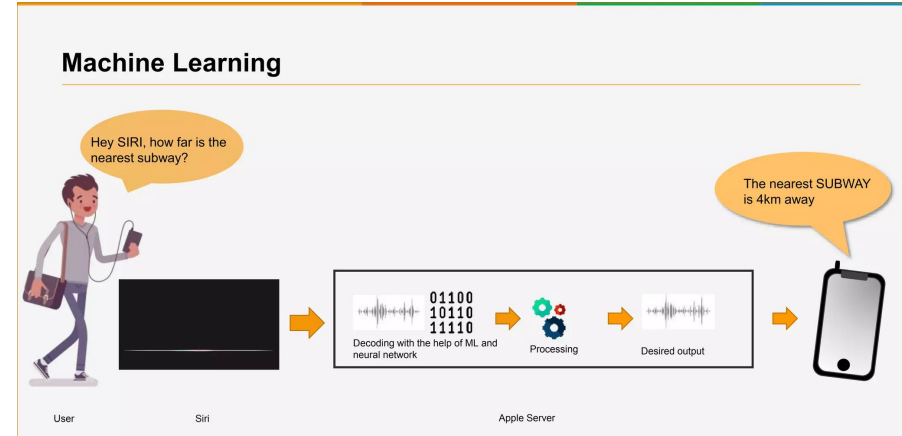
AI in Medical Field

- AI in the medical field has the potential to improve patient care, increase efficiency, accelerate research, and enhance the overall healthcare experience.
- It complements the expertise of healthcare professionals, leading to more informed decision-making and better health outcomes for patients.

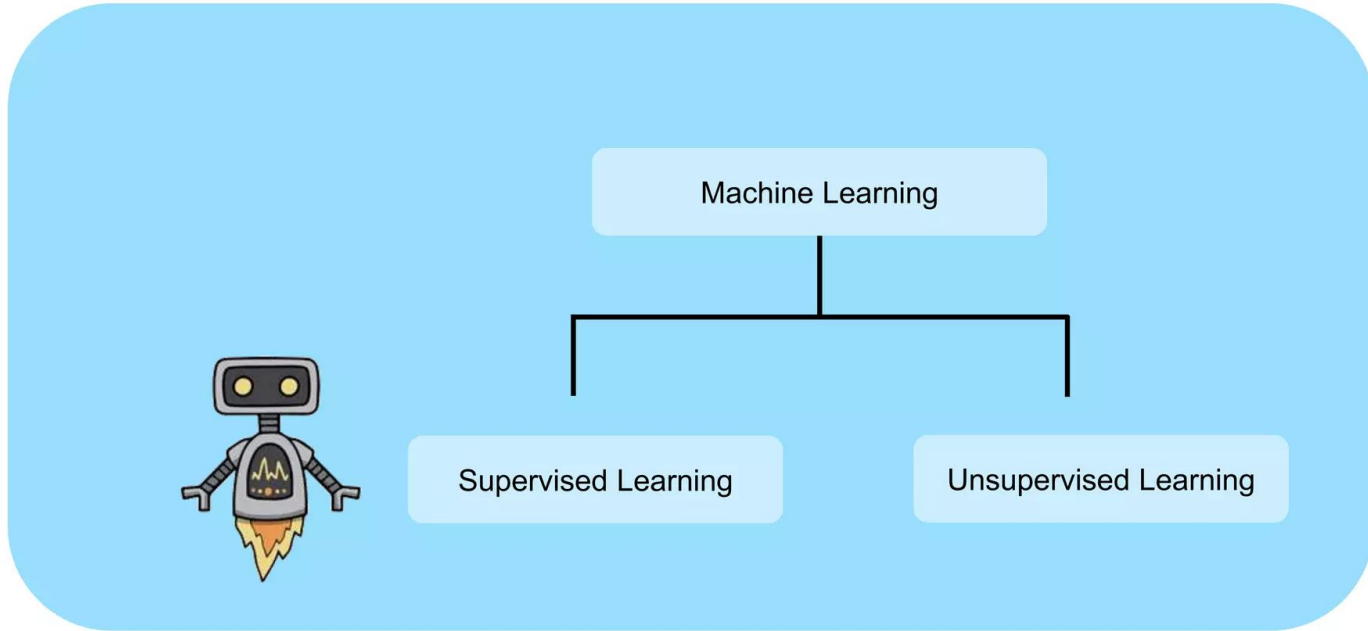


Artificial Intelligence Vs. Machine Learning

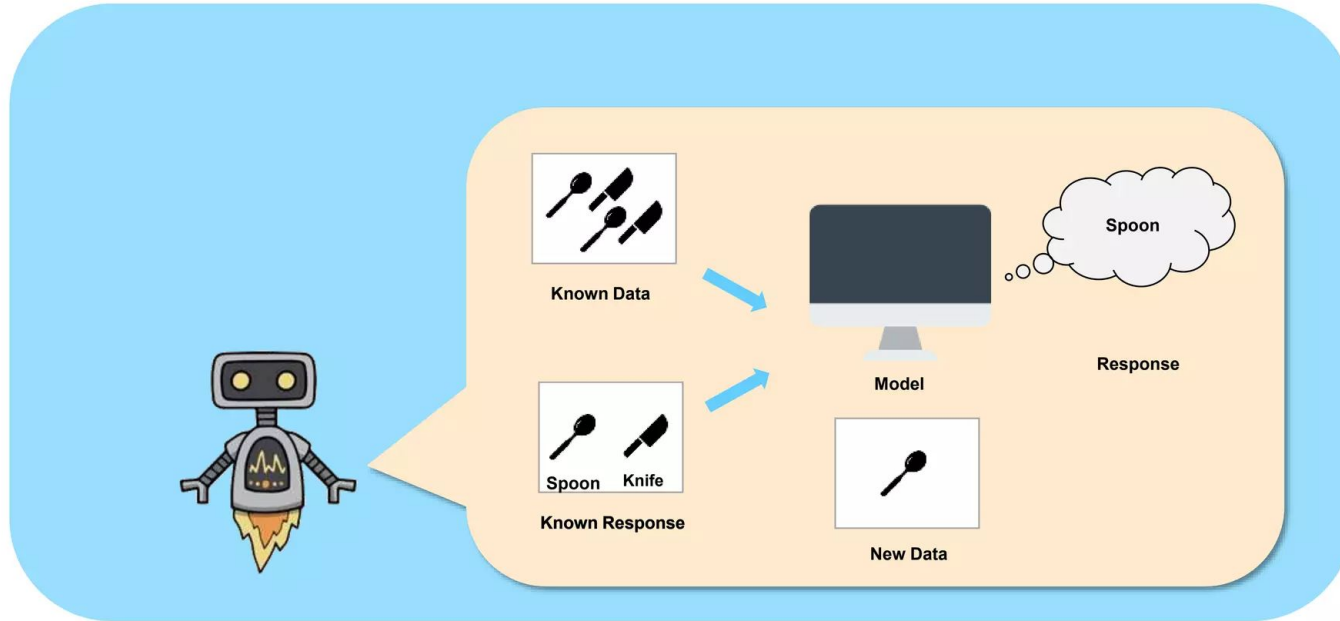
- **Machine Learning (ML):** A subset of AI, specifically involves the use of algorithms that allow machines to learn from and make predictions or decisions based on data.



Types of Machine Learning



Supervised Machine Learning



Supervised Machine Learning

Supervised Learning is basically of two types



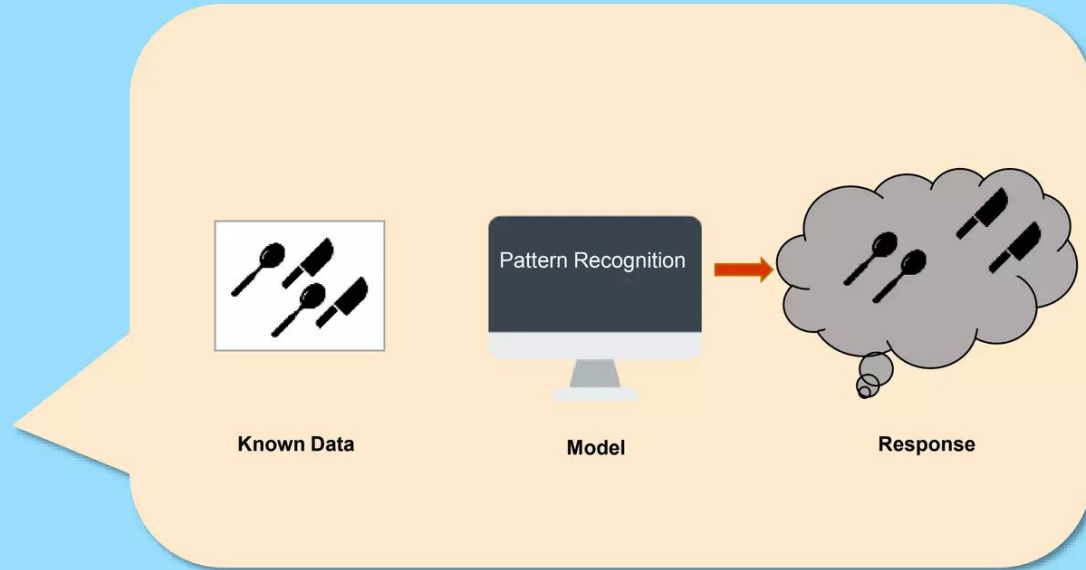
Classification

When the output variable is categorical i.e. with 2 or more classes (yes/no, true/false), we make use of classification

Regression

Relationship between two or more variables where a change in one variable is associated with a change in other variable

Unsupervised Machine Learning



Unsupervised Machine Learning

Unsupervised Learning is basically of two types



Clustering

The method of dividing the objects into clusters which are similar between them and are dissimilar to the objects belonging to another cluster

Association

Discovering the probability of the co-occurrence of items in a collection

Question 1

Can You Group the People in This Class?

- No predefined number of groups – create as many as you like.
- Use any criteria you think of to form the groups.
- Group sizes can vary - there are no restrictions.

Question 2

Is she part of Group A or Group B?
What do you mean by Group A & B?

I will provide some examples of known
people in these two groups,
which may help



Question 2

sample	Feature 1	Feature 2	...	Feature m	label
1	1	0.6	...	0.001	A
2	0	0.2	...	0.1	B
...	...	...	...	...	...
n	1	0.4	...	0.002	A
Our sample	1	0.5	...	0.005	??????

n = number of samples m= number of features

Question 2

Did the person attend the first session or not?

sample	Feature 1	Feature 2	...	Feature m	label
1	1	0.6	...	0.001	A
2	0	0.2	...	0.1	B
...	...	...	...	...	...
n	1	0.4	...	0.002	A
Our sample	1	0.5	...	0.005	??????

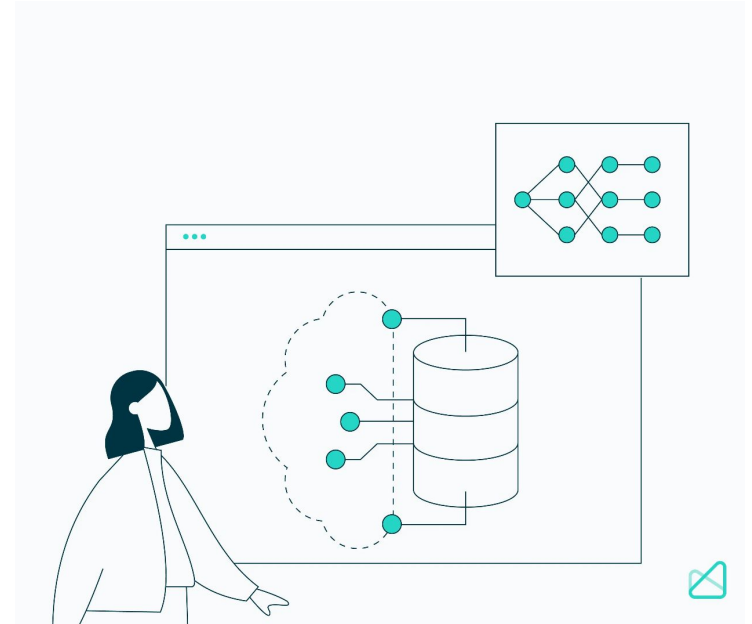
Non-medical student

Medical student

n = number of samples m= number of features

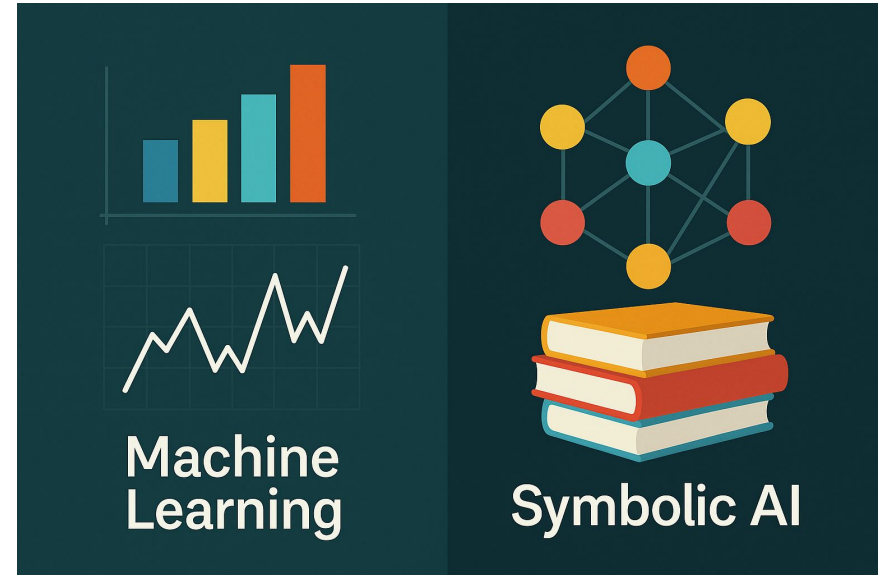
When ML Is Not Enough

- Requires large, high-quality labeled datasets, often hard to obtain in healthcare.
- Struggles with rare conditions or limited data populations.
- Lacks explicit reasoning, models learn patterns but can't explain why.
- May produce biased results if training data isn't representative.
- Difficult to incorporate clinical guidelines or expert rules directly.



From Patterns to Knowledge

- Machine Learning discovers patterns in data, but medicine also depends on knowledge, logic, and reasoning.
- To combine the strengths of both, AI can also be built on medical knowledge, rules, and relationships, this is Symbolic or Knowledge-Based AI.



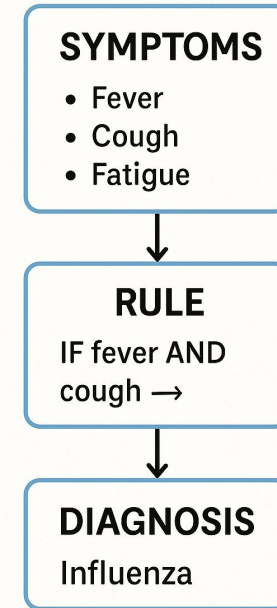
Symbolic (Knowledge-based) AI in Medicine

- Relies on explicit rules, ontologies, and expert-defined relationships.
- Mimics the way clinicians reason: "If symptom A + test result B → consider diagnosis C."
- Useful when data is limited but medical knowledge is well established.
- Forms the basis for clinical decision support systems (CDSS) and diagnostic reasoning tools.

Examples:

- Drug–drug interaction checkers
- Rule-based triage systems (e.g., early diagnosis)

MEDICAL REASONING

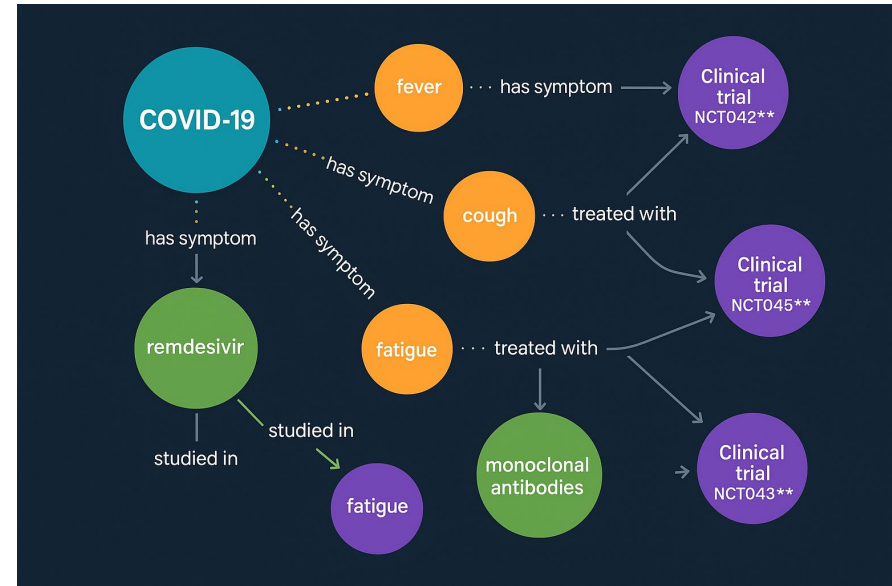


Knowledge Graphs in Healthcare

- Knowledge Graphs represent relationships between medical entities (diseases, symptoms, treatments, genes, etc).
- Enable reasoning, discovery, and integration across diverse data sources.
- Help link structured knowledge (e.g., guidelines) with unstructured data (e.g., notes, imaging reports).
- Support explainable AI by showing why a certain connection or recommendation was made.

Example:

- Linking COVID-19 → symptoms → treatments → clinical trials.

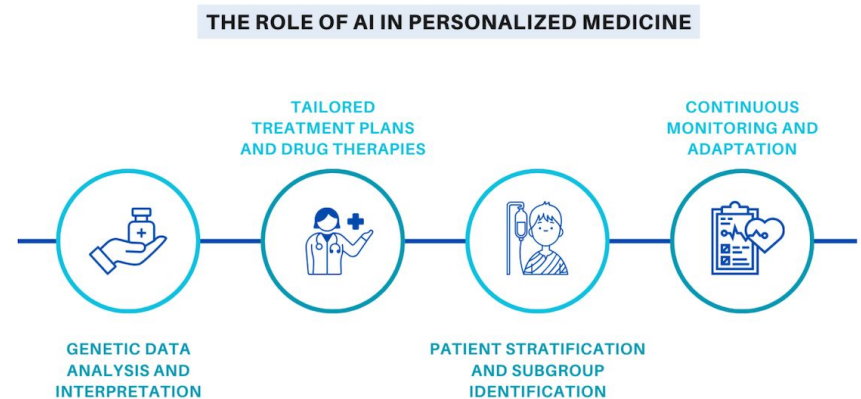


How Can AI Help

Application	Description
Diagnosis	AI algorithms can analyze medical images such as X-rays and 3D scans to detect abnormalities and make diagnoses
Predictive analytics	AI can analyze large amounts of data such as electronic medical records to predict the prospect of certain diseases and patient outcomes.
Personalized medicine	By analyzing a patient's medical and dental history and genetic makeup, AI can help create a personalized treatment plan
Drug discovery	AI can analyze compounds, predict their potential efficacy as drugs, and accelerate the drug discovery process
Clinical decision support	AI can provide physicians with real-time recommendations based on the latest evidence and guidelines

Personalized Medicine

- AI analyzes patient data including genetic information and medical history, to tailor treatment plans, medications, and therapies to individual patients, improving treatment outcomes.
- Algorithm designed takes input of patient, analyses, and gives medicine for disease.



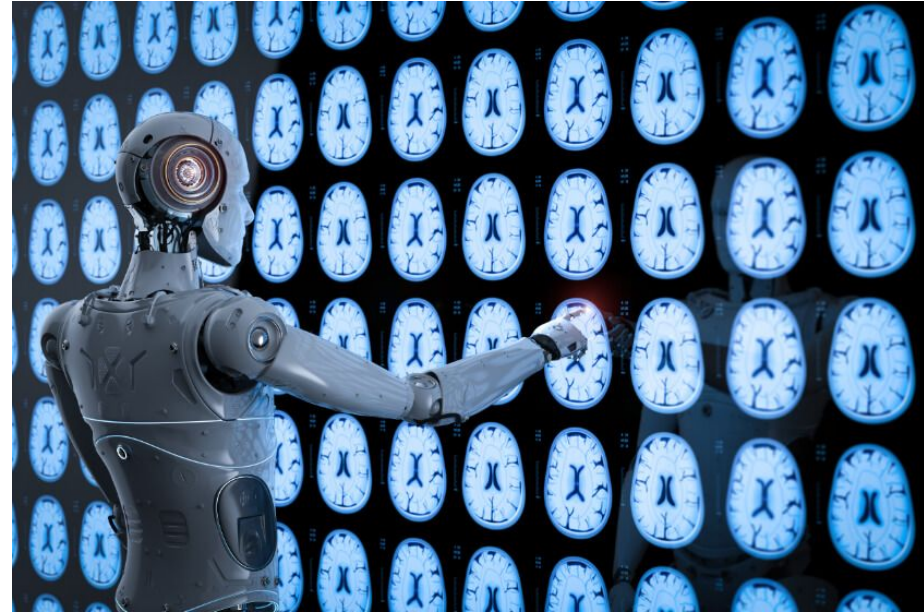
Diagnosis and Treatment

- AI algorithms can analyze medical data and help detect diseases and abnormalities quickly, aiding in early and accurate disease diagnosis.



Medical Imaging

- AI algorithms can analyze medical images (CT scans, MRI, X-rays), helping to detect diseases and anomalies with high accuracy. This speeds up diagnosis and reduces the risk of human error.



Robotic Surgery

- AI-assisted robotic surgery systems enhance the precision and capabilities of surgeons,
- leading to better surgical outcomes and reduced invasiveness.



Healthcare Chatbots

- AI-powered chatbots and virtual assistants provide support for patients, answering medical questions, scheduling appointments, and providing health information.
- AI-powered devices enable continuous remote patient monitoring, allowing healthcare providers to track patients' vital signs and health conditions in real-time, improving patient care and reducing hospital readmissions.



Genomic Medicine and Gene Mapping

- AI aids in the analysis of genetic data, helping identify genetic markers for diseases and potential treatment options.

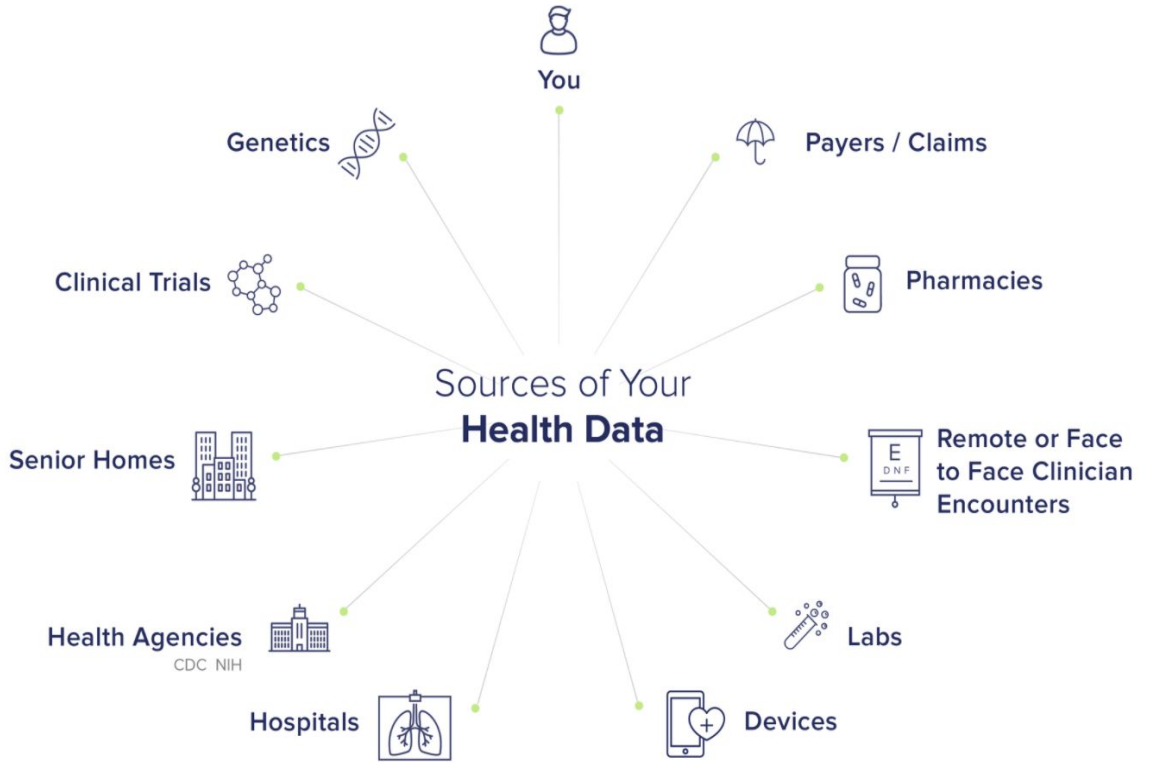


Drug Discovery

- AI can accelerate drug development by analyzing vast datasets to identify potential drug candidates and predict their effectiveness.



Where Does Health Data Come From



Data Sources

1- Primary data source

Team collects data to answer specific question.

PROS: Able to control quality of data collection and customize collection based on the features needed.

CONS: Very time consuming, more expensive, lots of ethics paperwork, extensive pre-processing steps required.

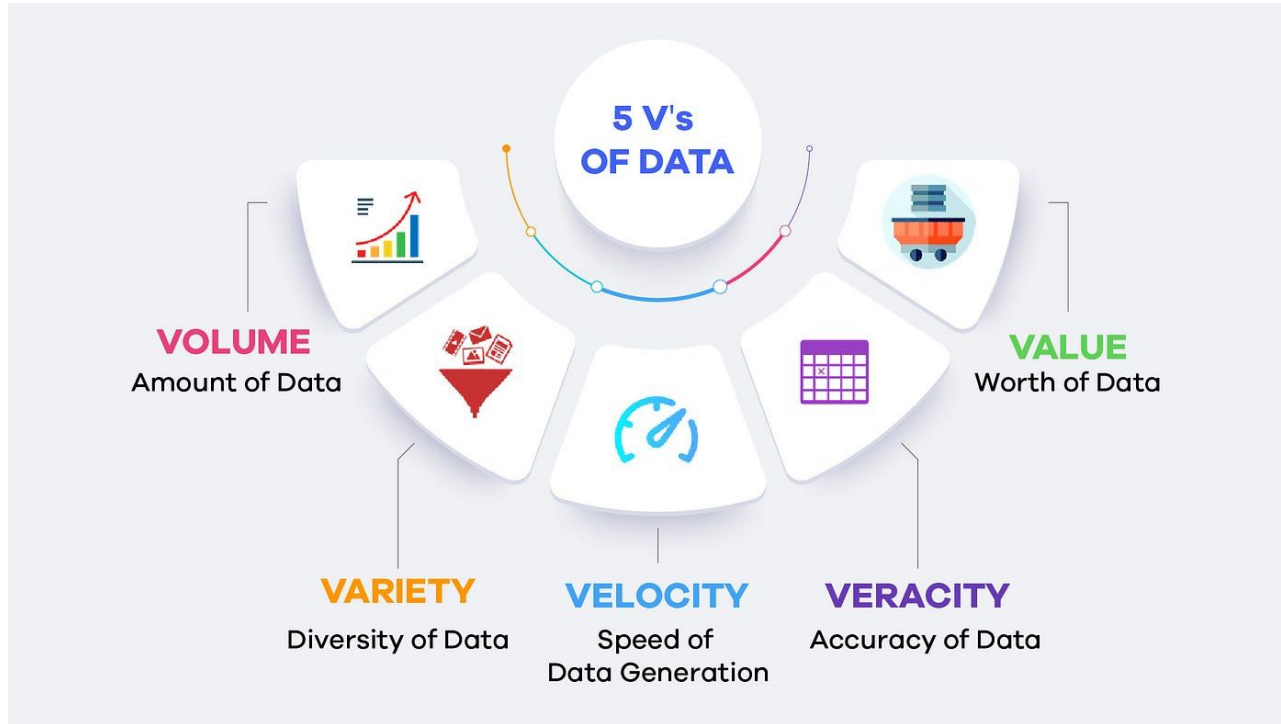
2- Secondary data source

Team requests data from a pre-existing database or study

PROS: Quick and cheap, able to gather large number of datapoints

CONS: May not have access to all the features you would want

5 V's of Big Data



=

Introduction to Health Data

Structured

Standardized

Easily transferable between health information systems

Ex: Age, gender, etc.

Unstructured

Not standardized

Significant pre-processing required prior to use

Ex: Radiographs (X-Rays and CBCT), scans, charting notes, etc.

Introduction to Health Data

Quantitative

Uses numbers

Easily transferable between health information systems

Ex: Age, weight, blood pressure, etc.

Qualitative

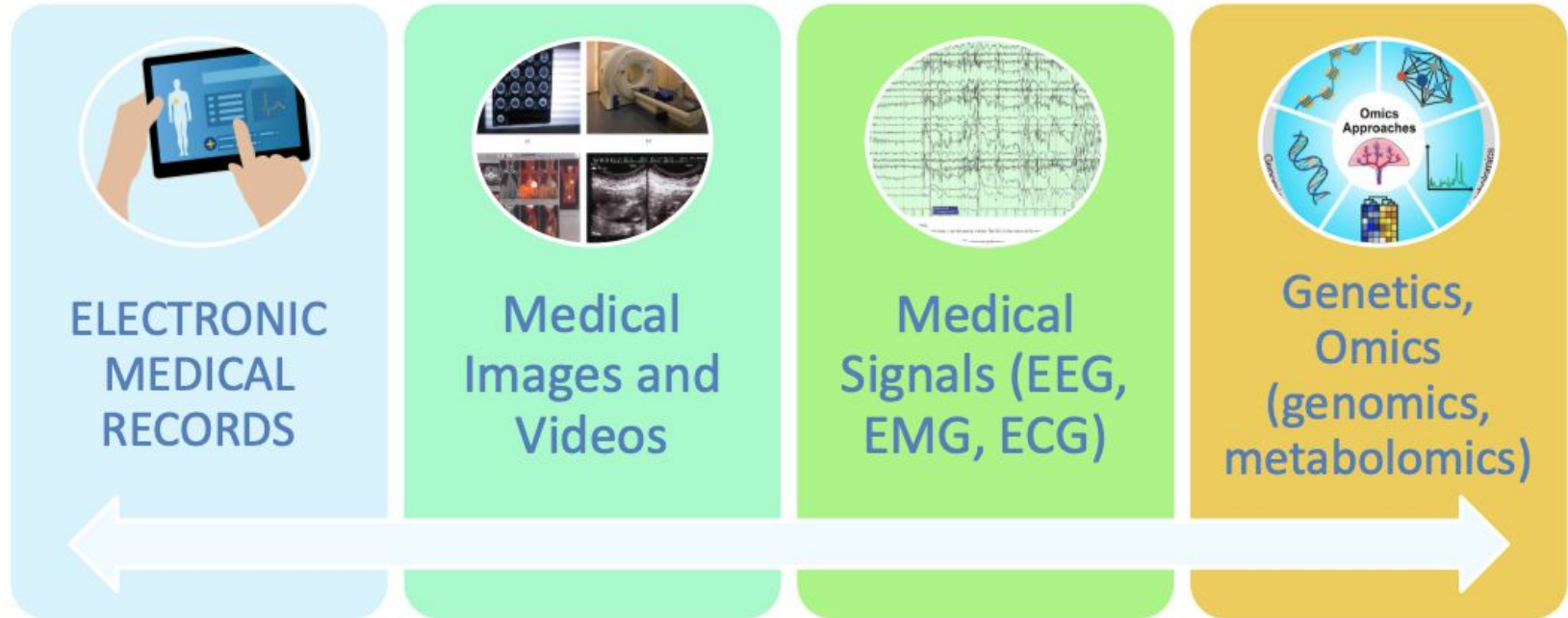
Non-numerical

Usually observational/descriptive

Transferability to different health systems depends on the data

“1730h: Mr. Smith presented with sharp headache”

Example of Different Types of Data in Medicine





Electronic Medical Records

EMR Introduction

Key idea: centralized and standardized health records

Use of computers in medical practices in the United States has increased gradually over the past decades. In 1983–84, approximately 19.7% of family physicians reported using a computer in their office, with around 14.0% owning or leasing one.

Patient Chart [000006 COLEMAN, AMBER]

File Daily Export Lists Pt Chart Tasks Templates Encounters Rx Image WP Modules Help

Patient # 000006 SSN 666993333 Last Name COLEMAN First Name AMBER MI Chart No

Address 680 SYCAMORE ST Status Active Pt Type PT - PATIENT Sig Date 05/22/2011

DOB 08/16/1973 Age 47 yrs Provider DAVID MILLER

Home Ph 618-772-8446 Cell Ph 618-772-8446 Sex FEMALE Occupation INVENTORY MGR

Work Ph 618-772-8446 Ext Marital Status MARRIED Pharmacy 0002 CVS PHARMACY

Employer KOHL'S DEPT STORE Recall Dt Email Address spattton@americanmedical.com

Last Note 04/26/2016

Next Appointment

Immunizations Scheduled RHCM Items Due Allergy Alert

Encounter Notes Active

Encounter	Date	Notes
0000000074	04/26/2016	PSYCH VISIT
0000000073	04/22/2016	CARDIO VISIT
0000000072	04/20/2016	2 MO WCC
0000000071	04/04/2016	WELL WOMAN

Test Tracking

Test	Date	Result
0000000190	02/19/2016	02/19/2016 AN
0000000191	02/19/2016	02/19/2016 HEF
0000000192	02/19/2016	02/19/2016 HIV
0000000193	02/19/2016	02/19/2016 HC

Medications

Medication	Dose	Frequency	Start Date
ZITHROMAX 250 MG...	TAKE ...	11/22/2016	
SUBOXONE 8 MG-2 ...	TAKE ...	08/02/2016	
LASIX 40 MG TAB		10/02/2014	
ZITHROMAX 250 MG...	TAKE ...	06/03/2014	

Images

Image	Date	Description
000010	08/03/2016	test
000009	04/13/2016	* asdfasf
000008	03/30/2016	patient pic
000007	11/11/2015	puppy

Print Encounter Note Entry Measures / Alerts Tracking Entry Supplementary Clear

Task Search Enter/Update Vitals C32 / CCR Documents Order Tests Write Rx Exit

What Does EMR Data Include



Advantages of EMR Data

- Both broad and deep
 - Contains thorough detail about an individual
 - Does this for (often) millions of patients!
- Deep learning often works best with this size of data
- This data can be used to learn both about individuals, populations, healthcare operations management, and more
- The data itself is a byproduct, we aren't spending extra money to generate it for research purposes

Drawbacks of EMR Data

- Completeness/quality/integrity issues
- Incredibly sensitive information
- Sheer scale
- Many different data types
- Longitudinal complexity

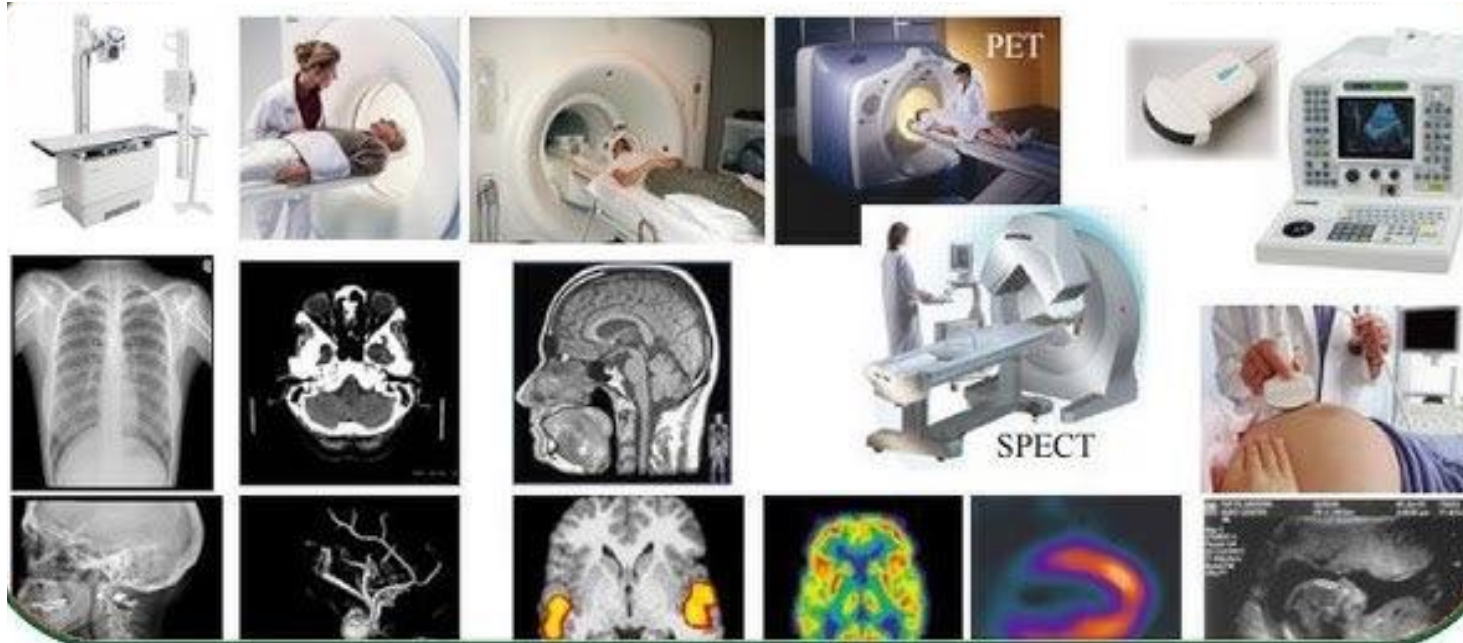


Medical Imaging

Introduction

- With recent advancements in medical imaging technology, clinicians can now capture highly detailed anatomical and functional information at both micro and macro levels, greatly improving disease detection, diagnosis, and treatment planning.
- **Medical Imaging:** Refers to the use of advanced imaging modalities such as X-ray, CT, MRI, ultrasound, and PET to visualize internal body structures for diagnosis, treatment planning, and monitoring of medical conditions.

Types of Medical Imaging



Advantages of Medical Imaging Data

Higher Diagnostic Accuracy

Digital imaging enhances visualization of anatomical structures, supporting early detection of diseases and abnormalities.

Enhanced Treatment Planning

3D imaging modalities such as CT and MRI provide detailed anatomical information for precise surgical and interventional planning.

Easier Data Retrieval & Sharing

Cloud-based PACS and EMR systems enable instant access to previous images and seamless sharing among healthcare providers.

Lower Radiation Exposure

Digital imaging technologies reduce radiation dose compared to conventional film-based methods.

AI-Assisted Analysis

Machine learning algorithms can automate image interpretation tasks such as lesion detection, organ segmentation, and disease classification.

Disadvantages of Medical Imaging Data

Large File Sizes

High-resolution CT, MRI, and PET images require substantial storage space and may slow retrieval times.

High Initial Investment

Advanced imaging equipment such as MRI, CT, and PET scanners are expensive to purchase and maintain.

Training & Software Dependency

Accurate interpretation and workflow efficiency depend on specialized training and advanced imaging software.

Image Artifacts & Errors

Motion, metal implants, or technical issues can produce artifacts that affect image quality and diagnostic accuracy.

Privacy & Security Risks

Cloud-based image storage and PACS systems may be vulnerable to data breaches if not properly encrypted or managed.



Charting and Clinical Notes

Introduction

- Charting and clinical documentation are essential for recording patient history, diagnosis, and treatment plans in modern healthcare.
- Electronic Medical Record (EMR) systems have replaced traditional paper charts, improving data structure, accessibility, and clinical efficiency.
- These systems capture vital signs, diagnostic results, medication lists, and progress notes, while clinical documentation details diagnoses, interventions, and follow-up care.
- Advanced features such as speech-to-text entry and AI-assisted documentation enhance accuracy, reduce clinician workload, and streamline clinical workflows.

Types of Charting and Clinical Notes

Includes vital sign charting, medication records, progress notes, diagnostic test reports, imaging annotations, and voice-to-text transcriptions each providing detailed insights into different aspects of patient care and clinical management.

EMERGENCY DEPARTMENT HISTORY AND PHYSICAL EXAMINATION

DATE (YYYY/MM/DD)	TIME (HH:MM)	SIGNATURE AND PRINTED NAME
2020/01/01	14:00	25F w/ abdo pain x 1hr.
		- RLQ pain, non-radiating, cov
		worsening. Tried Tylenol 101
		Q/N/V x 3 episodes. Last meal
		normal BM - ϕ diarrhea. ϕ
		ϕ flatulence ϕ surgical hx
		- ϕ dysuria ϕ urinary frequency
		- ϕ abnormal vaginal discharge
		periods regular. Sexually active
		inconsistently. Last intercourse
		O/E T 37°C HR (110) RR 16
		Gen: uncomfortable, lying
		CVS: S ₁ + S ₂ , diminished S ₄
		Abdo: guarding, RLQ tend
		tenderness. ϕ shute to
		A/P 25F w/ severe RLQ pain, 1
		ovarian torsion, UTI
		1) Abdo/pelvic U/S
		2) CBC, lytes, Cr, LFTs, lipase.
		3) Urinalysis
	16:50	R/A: pain improved. ϕ further
		OE: appears more comfortable
		lytes: lytes (2.2) LFTs + b1
	18:30	R/A: U/S shows appendiceal
		enlargement, likely appendicitis.
		uterus ovaries/Pelvic tubes unremarkable
		A/P: Appendicitis - Dispo: consult General Surgery for admission/surgery
		- continue NPO + Ringer's lactate 100cc/h maintenance

MEDICAL INFORMATION

NAME:	DOB:	SSN:
ALLERGIES:		
PRIMARY DOCTOR:		PATIENT HEALTH SYSTEM INFO:
NAME:	PATIENT ID:	
COMPANY NAME:	PASSWORD:	
PHONE:	RECORD NUMBER:	
FAX:	EMAIL:	
ADDRESS:		
AFTER HOURS NURSE:		
HEALTH INSURANCE: ☐ HMO ☐ PPO		EMERGENCY CONTACT #1:
COMPANY NAME:	NAME:	
GROUP NUMBER:	PHONE:	
HOLDER:	ALT. PHONE:	
SSN:	ADDRESS:	
CLAIMS ADDRESS:	RELATIONSHIP:	
PHONE:		
FAX:		
SPECIALIST:		EMERGENCY CONTACT #2:
NAME:	NAME:	
COMPANY NAME:	PHONE:	
PHONE:	ALT. PHONE:	
FAX:	ADDRESS:	
ADDRESS:	RELATIONSHIP:	
AFTER HOURS NURSE:		
SPECIALIST:		CURRENT MEDICATIONS:
NAME:		
COMPANY NAME:		
PHONE:		
FAX:		
ADDRESS:		
AFTER HOURS NURSE:		

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Advantages of Charting and Clinical Notes

Standardized Documentation

Digital charting eliminates illegible handwritten notes and ensures consistency.

Improved Treatment Tracking

Provides an organized history of treatments and periodontal status.

Easier Integration with AI & Decision Support

AI-assisted notes can flag missing data and suggest treatments.

Faster Input Methods

Speech-to-text technology can speed up note-taking.

Secure & Remote Access

Cloud-based records allow dentists to review patient charts from multiple locations.

Disadvantages of Charting and Clinical Notes

Training and Adaptation

Transitioning from paper-based to digital notes may require training.

Speech Recognition Errors

Voice-to-text systems may misinterpret medical terminology, leading to inaccurate records.

Ethical Issues

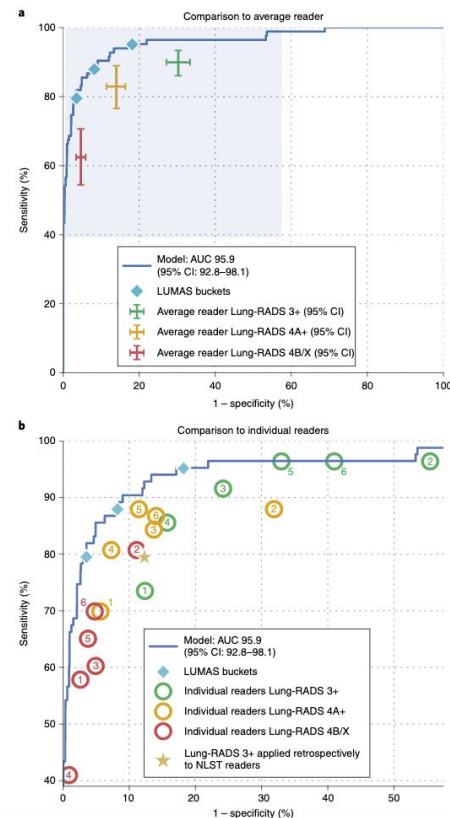
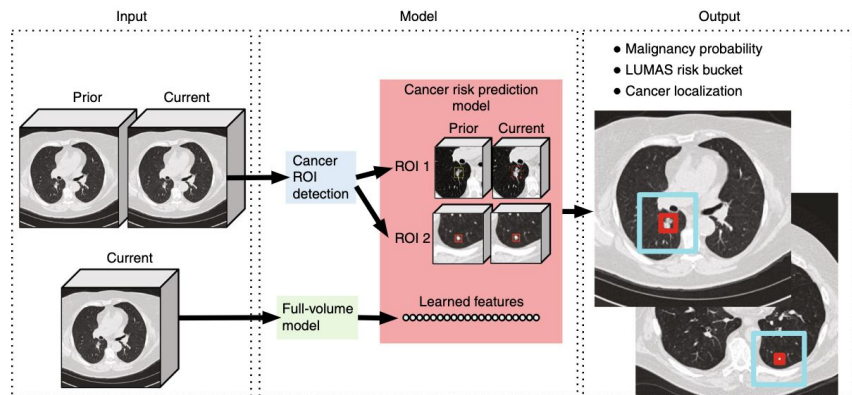
Digital records are susceptible to hacking if not properly encrypted.

Incomplete or Inaccurate Entries

Errors in manual entry or auto-generated notes can impact diagnosis and treatment.

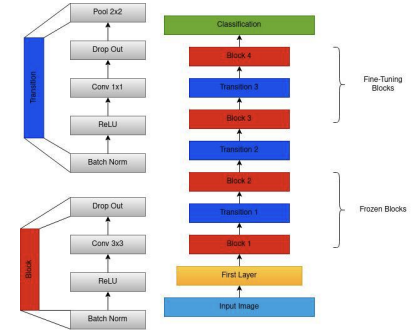
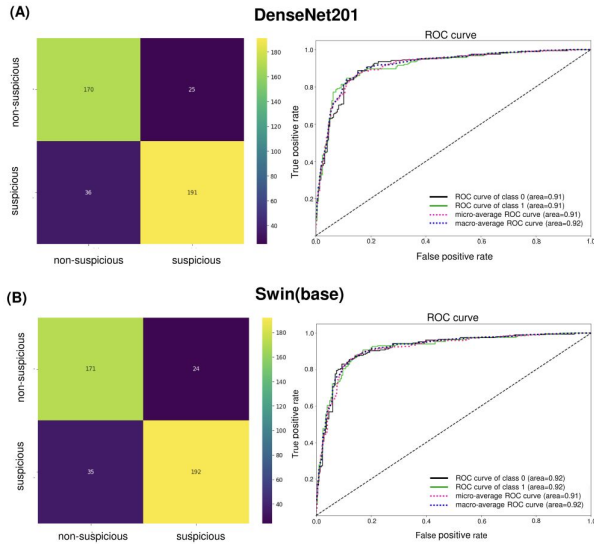
Screening

End-to-end lung cancer screening with three-dimensional deep learning on low-dose chest computed tomography (chest CT) ([Ardila et al, 2019](#))



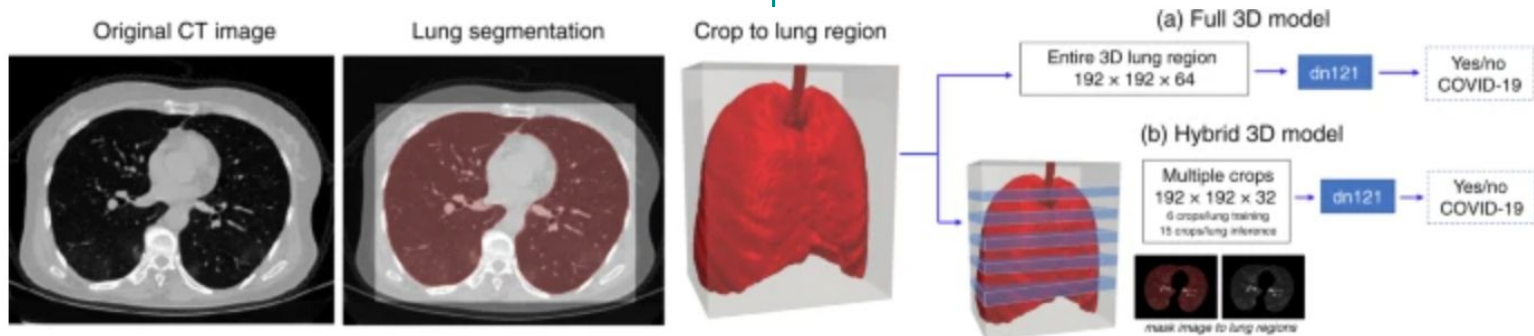
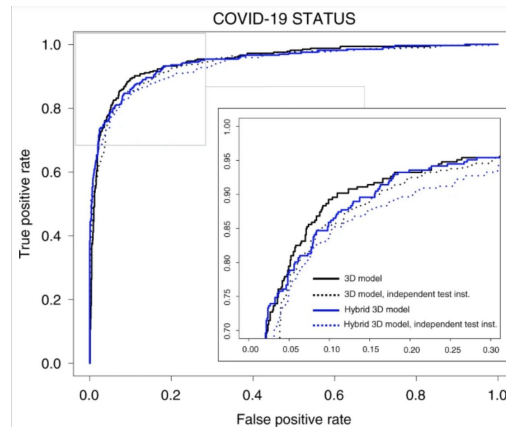
Screening

AI-Assisted Screening of Oral Potentially Malignant Disorders Using Smartphone-Based Photographic Images (Talwar et al., 2023)



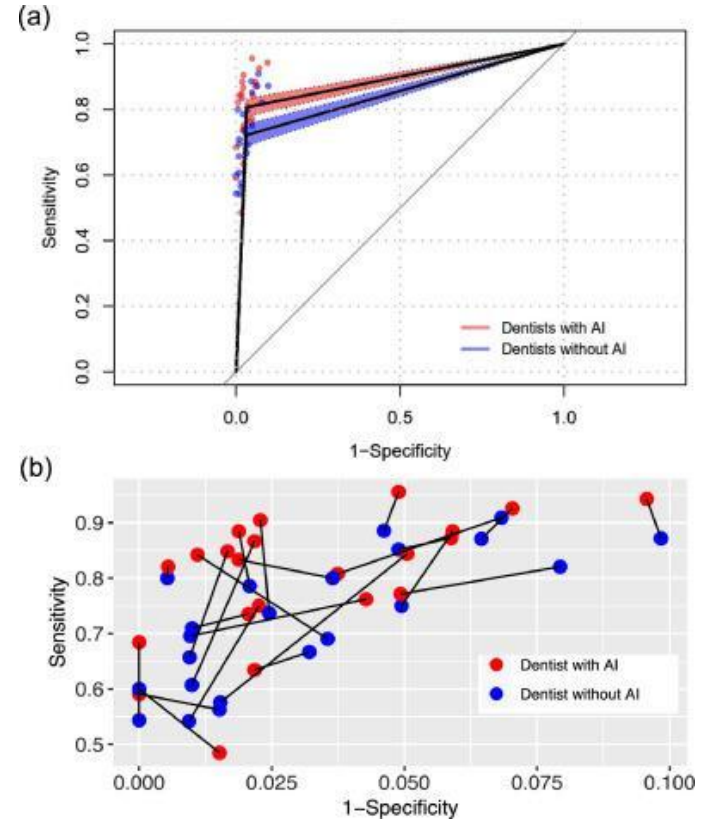
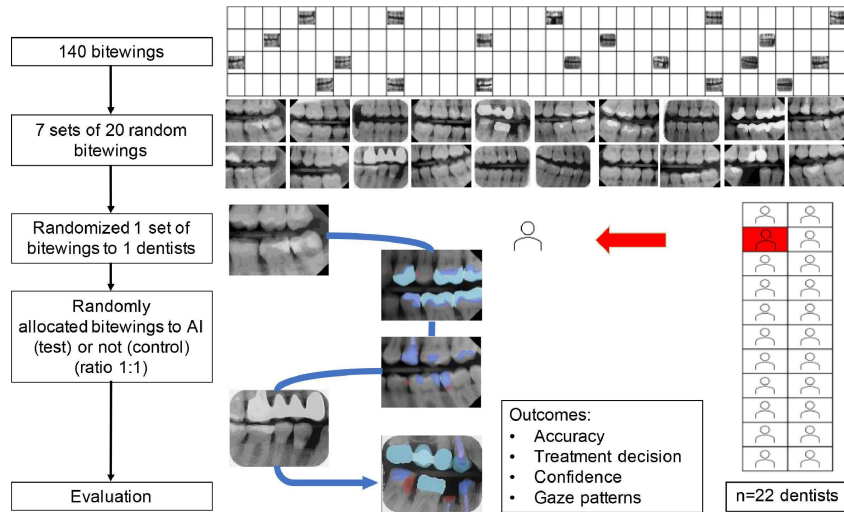
Diagnosis

Artificial intelligence for the detection of COVID-19 pneumonia on chest CT using multinational datasets ([Harmon et al., 2020](#))



Diagnosis

Artificial intelligence for caries detection:
Randomized trial (Mertens et al., 2021)





Benefits and Challenges of AI Integration

Advantages of AI in Enhancing Patient Care

Improved Diagnostic Precision

AI technologies enhance diagnostic precision by utilizing advanced algorithms that analyze imaging data, leading to earlier detection of health issues and more effective treatment strategies tailored to individual patient needs.

Proactive Patient Management

Through predictive analytics, AI identifies potential health risks, allowing health professionals to implement preventive measures that reduce the likelihood of severe conditions and associated treatment costs.

Enhance Patient Accessibility

AI-driven tele-medicine expands access to care, enabling remote consultations and follow-ups, particularly beneficial for patients in rural or underserved areas, thus improving overall patient engagement and satisfaction.

Potential Obstacles to AI Adoption in Medicine

Data Privacy Regulations

Compliance with strict healthcare data privacy laws, such as HIPAA and GDPR, is critical for medical institutions. Failure to safeguard patient data can lead to legal implications, breaches of confidentiality, and loss of public trust, significantly slowing AI implementation in healthcare.

Algorithm Bias Risk

AI systems may exhibit biases if trained on non-representative datasets, leading to inaccurate diagnoses. Continuous monitoring and diverse data inclusion are necessary to ensure equitable and safe AI applications in healthcare.

High Initial Investment

The financial burden of acquiring AI technologies, including software and training, can deter smaller practices. A clear demonstration of long-term benefits and ROI is crucial to encourage adoption among these providers.

Thank You!

Stay in Touch

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